Finance Research Letters 67 (2024) 105688



Contents lists available at ScienceDirect

Finance Research Letters

journal homepage: www.elsevier.com/locate/frl





Revolutionizing healthcare: Unveiling China's healthcare efficiency quest and path to progress

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ARTICLE INFO

Keywords:
Social healthcare expenditure
Government health investment
Investment efficiency
Resource duplication

ABSTRACT

Social healthcare expenditure is a crucial component of global healthcare expenditure. Unlike government investment, the efficiency of social-level investment encounters more problems and challenges. Using China as an example, this study examines the efficiency of both social and governmental healthcare investments, revealing the primary challenges facing social healthcare expenditure. The findings indicate that despite significant investment growth, inefficiencies persist due to resource duplication and misallocation. Additionally, declining technological progress has resulted in lower productivity, underscoring the urgent need for innovation. This study emphasizes the necessity of accelerating technological advancements to enhance the efficiency of social healthcare investments, improve resource utilization, and ensure the equitable distribution of healthcare resources nationwide.

1. Introduction

China's 14th Five-Year Plan prioritizes a comprehensive overhaul of healthcare performance. While government investment in healthcare continues to rise, long-term growth in expenditure faces inevitable constraints. To address the persistent issue of high medical costs, it is crucial to simultaneously enhance investment in healthcare at the societal level. Existing research, notably by Flessa (2007), Kerr and Hendrie (2018), and Sun et al. (2024), has provided valuable insights into healthcare investment efficiency. However, there remains a significant lack of comprehensive investigations into the societal dimension, as emphasized by Hale et al. (2020), Hong et al. (2024), Lacy-Nichols et al. (2023), Qehaja et al. (2023), and Shahid et al. (2024). Therefore, the government must create a platform to effectively utilize social healthcare as a safety net.

This study employs Data Envelopment Analysis (DEA) and the Malmquist index method to analyze the efficiency of current government and social healthcare investment expenditures in China. The DEA method is chosen for its suitability in systems with

Funding information: Doctoral Science Research Program in Xichang University "Research on the Influence of Job Insecurity on Enterprise Knowledge Workers' Innovative Behavior" (Grant/Award Number: YBS202321)

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https://doi.org/10.1016/j.frl.2024.105688
Received 24 April 2024; Received in revised form 27 May 2024; Accepted 31 May 2024
Available online 2 June 2024
1544-6123/© 2024 Published by Elsevier Inc.

multiple inputs and outputs, as it eliminates the need for estimating production functions and effectively addresses the challenges associated with processing dispersed indicators using global data. However, traditional CCR and BCC models only reflect static efficiency and lack comparability across different periods. In contrast, the Malmquist productivity index, which can be decomposed into technical efficiency, technological progress, and scale efficiency, offers significant advantages in assessing total factor productivity across Chinese provincial regions. The DEA-Malmquist model index not only addresses operational performance but also can be widely applied across various fields, revealing productivity changes through improvements in efficiency and technological advancements.

2. Data and methods

2.1. Data collection

The data utilized in this study were extracted from the "China Health Statistics Yearbook" (formerly the "China Health and Family Planning Statistics Yearbook") spanning the years 2012 to 2021. This annual publication, issued by the National Health Commission of the People's Republic of China, offers a comprehensive overview of the development of China's health sector and the health status of its residents. Given the variability in the criteria for calculating healthcare investments and the unavailability of data for Hong Kong, Macao, and Taiwan, this study focuses exclusively on data from the 31 provinces and municipalities in mainland China.

2.2. Analytical framework

The analysis employs panel data from 2012 to 2019 for the 31 mainland provinces, utilizing the DEA-Malmquist index model to evaluate efficiency. The primary objective is to compute the total factor productivity (TFP) index and the geometric mean of its decomposition components. These computations are performed using DEAP 2.1 software. The chosen indicators represent average values over the eight-year period under study.

2.3. Variable selection

In selecting the input and output indicators, this study draws upon a substantial body of domestic and international studies on the assessment of the efficiency of healthcare inputs and outputs (Baracos et al., 2018). In order to ensure a comprehensive analysis that encompasses both governmental and social dimensions (Tanimura et al., 2014), this study incorporates input and output indicators in order to construct a new indicator system, as shown in Table 1. The output indicators provide a comprehensive measure of the efficiency of the availability and utilization of healthcare services across the country (Selden and Berdahl, 2020). The combination of these input and output indicators allows this study to provide a comprehensive assessment of the efficiency of China's healthcare investments. This assessment identifies potential areas for improvement and ensures better healthcare outcomes.

2.4. Model

2.4.1. Data envelopment analysis

The DEA models offer distinctive benefits in evaluating efficiency, particularly in intricate multi-input, multi-output settings. However, their practical implementation necessitates rigorous control of data quality and a consideration of the assumptions and constraints inherent to the model when interpreting the outcomes. The DEA model evaluates relative efficiency by maintaining decision-making units' inputs or outputs unchanged and identifying the efficient production frontier. Units are projected onto this frontier, and their relative efficiency is measured by assessing deviations from it. The basic DEA models include the CCR and BCC models. The CCR model assumes constant returns to scale, while the BCC model allows for variable returns. Both models assess units for DEA efficiency. The CCR model involves non-Archimedean infinitesimal quantities, while the BCC model represents these by ε.

Table 1
Established indicator system.

System of indicators	Type of indicator	Indicator name
System for evaluating the effectiveness of health cost inputs and outputs	Indicators in the category of government inputs	Government health expenditure
		Per capita health costs
	Indicators in the category of social inputs	Social health expenditure
		Per capita health costs
	Output category indicators	Number of health institutions
		Number of beds in medical institutions
		Number of personnel in the Centre for Disease Control and Prevention
		Number of staff in specialized public health institutions
		Health technicians per 1000 population

$$\min\left[\theta-\varepsilon\left(\sum_{j=1}^{m}s^{-}+\sum_{j=1}^{r}s^{+}\right)\right]$$
$$s.t.\begin{cases}\sum_{j=1}^{n}x_j\lambda_j+s^{-}=\theta x_0\\ \sum_{j=1}^{n}y_j\lambda_j-s^{+}=y_0\\ \lambda_j\geqslant 0\\ s^{+}\geqslant 0,:\geqslant 0\end{cases}\tag{1}$$

$$\min\left[\theta-\varepsilon\left(\sum_{j=1}^{m}s^{-}+\sum_{j=1}^{r}s^{+}\right)\right]$$
$$s.t.\begin{cases}\sum_{j=1}^{n}x_j\lambda_j+s^{-}=\theta x_0\\ \sum_{j=1}^{n}y_j\lambda_j-s^{+}=y_0\\ \lambda_j\geqslant 0\\ s^{+}\geqslant 0,s^{-}\geqslant 0\\ \sum_{i=1}^{n}\lambda_i=1\end{cases}\tag{2}$$

The Malmquist index, commonly applied to panel data, provides deeper insights into dynamic efficiency changes compared to traditional CCR and BCC models, mainly suited for single-time-point efficiency comparisons. By capturing dynamic efficiency shifts, the Malmquist index is adept at analyzing historical data across regions or industries. It operates via distance functions, defining the distance function using directional outputs and inputs. The distance function for output variables is typically expressed as:

$$D_0(x,y)=\inf\{\delta:(x,y)/\delta\in p(x)\}\tag{3}$$

The Malmquist productivity index can be expressed as follows:

$$M_0(x_{t+1},y_{t+1},x_t,y_t)=\left[\frac{D_0^t(x_{t+1},y_{t+1})}{D_0^t(y_t,x_t)}\times\frac{D_0^{t+1}(x_{t+1},y_{t+1})}{D_0^{t+1}(y_t,x_t)}\right]^{1/2}\tag{4}$$

Research scholars have further decomposed and refined the Malmquist index. The calculation of the Malmquist productivity index requires the computation of four distance functions:$D_0^t(x_t,y_t),\quad D_0^{t+1}(x_{t+1},y_{t+1}),\quad D_0^t(x_{t+1},y_{t+1})$和$D_0^{t+1}(x_t,y_t)$.

2.4.2. Total factor productivity index

The Total Factor Productivity Index (TFPCH) is utilized to measure the efficiency of government and social inputs and outputs in healthcare. TFPCH reflects the "efficiency of production activities over a certain period" and serves as a productivity indicator, quantifying the ratio of total output to total input factors. TFPCH originates from three main sources: (1) technological progress, (2) efficiency improvement, and (3) scale effects. Consequently, TFPCH can be decomposed into the Technological Progress Index (TECHCH) and the Composite Technical Efficiency Change Index (EFFCH). The EFFCH represents the extent to which existing technology is effectively utilized. Under the assumption of constant returns to scale, EFFCH can be further decomposed into the Scale Efficiency Change Index (SECH) and the Pure Technical Efficiency Change Index (PECH). SECH indicates whether changes in the increments of medical output and input are consistent, while PECH provides deeper insights into changes in the management level of medical and health institutions. The decomposition of the Malmquist index can be expressed by the following formula:

$$\text{TFPCH}=\text{TECHCH}\times\text{EFFCH}=\text{TECHCH}\times\text{SECH}\times\text{PECH}\tag{5}$$

When the TFPCH result is greater than or equal to 1, it signifies that healthcare expenditure is effective, indicating a relatively high overall performance level of healthcare investment. Conversely, when the TFPCH result is less than 1, it denotes that healthcare expenditure is ineffective, suggesting a relatively low overall performance level of healthcare investment.

3. Results

3.1. Trends in healthcare expenditure

As shown in Table 2, from 2012 to 2019, society's healthcare expenditure increased significantly, from 1003.070 billion yuan to 2915.057 billion yuan, with an average annual growth rate of 16.3 %, while government expenditure also rose from 843.198 billion yuan to 1801.695 billion yuan, with an average annual growth rate of 10.3 %. Clearly, social healthcare expenditure grew significantly faster than government expenditure, reflecting a substantial increase in the participation and contribution of social capital in the healthcare sector. Per capita healthcare costs also increased from 2068.8 yuan in 2012 to 4702.8 yuan in 2019, an increase of about 2.27 times, indicating an overall improvement in healthcare services and costs.

At the social level, the rapid growth in healthcare expenditure has mainly been driven by the development of private healthcare institutions, the expansion of the insurance industry, and the increased emphasis on health investment by the community. The absolute value of social expenditure has increased by about 2.91 times, far exceeding the 2.14-fold increase in government expenditure, suggesting that social capital has been increasingly invested in healthcare resources, leading to greater diversification and quality of healthcare services (Inhorn and Patrizio, 2015). The rise in healthcare expenditure at the social level, including private hospitals, private medical institutions, social insurance, and personal health consumption, has supplemented and optimized the allocation of public healthcare resources (Atun et al., 2015)."

Overall, the rapid growth in social healthcare expenditure and the significant rise in per capita healthcare costs reflect the important role and influence of social capital in the healthcare sector. With increasing social investment, the supply and quality of healthcare services are expected to improve further to meet the growing demand for healthcare (Marrast et al., 2016). This also suggests that the State needs to pay attention to the fairness and efficiency of the distribution of social healthcare resources to ensure the continued improvement of health for all (Jakovljevic et al., 2023).

3.2. Comparing government and societal efficiency

Unlike the government's performance, the efficiency of society's investment in healthcare and technological progress has fluctuated considerably. However, overall efficiency has improved, particularly in terms of pure technical efficiency and efficiency of scale. The efficiency of input-output is evaluated by examining government and social healthcare investments as decision-making units (Kamper et al., 2014).

According to the data in Table 3, the indicators EFFCH (change in efficiency), TECHCH (change in technology), PECH (change in pure technical efficiency), SECH (change in scale efficiency), and TFPCH (change in total factor productivity) reflect the efficiency of government and societal inputs as well as technological progress in healthcare. From 2012 to 2019, the government's EFFCH indicator was above 1.0 for most of the period, indicating an improvement in efficiency, but it declined to 0.946 and 0.976 during 2016–2017 and 2018–2019, respectively, showing some volatility. In contrast, society's EFFCH indicator was also above 1.0 overall, reaching 1.044 and 1.089 during 2012–2013 and 2017–2018, respectively, indicating more significant efficiency gains.

The government's TECHCH indicator peaked at 1.066 in 2016–2017 but mostly hovered around 0.9 in other periods, indicating fluctuations in technological progress. Society's TECHCH indicator also peaked at 1.026 in 2016–2017 but was generally lower than the government's, particularly in 2017–2018 and 2018–2019, at 0.798 and 0.443, respectively, showing large fluctuations in technical progress.

Regarding pure technical efficiency change (PECH), the government's indicator was close to 1.0 overall, indicating a relatively stable change in pure technical efficiency, but it declined to 0.965 and 0.977 in 2016–2017 and 2018–2019, respectively. In contrast, society's PECH indicator was above 1.0 in 2013–2014 and 2014–2015, showing some pure technical efficiency gains.

For Scale Efficiency Changes (SECH), the government's SECH indicator peaked at 1.038 in 2012–2013 but then fluctuated. Society's SECH indicator was 1.056 and 1.073 in 2012–2013 and 2017–2018, respectively, showing significant scale efficiency improvements.

In terms of Total Factor Productivity Change (TFPCH), the government's indicator reached its highest value of 1.009 in 2016–2017 but generally fluctuated. Particularly notable were the values of 0.898 in 2017–2018 and 0.439 in 2018–2019. Conversely, society's TFPCH indicator was generally below 1.0, indicating a significant decline in total factor productivity, especially in 2018–2019 when it reached its lowest value of 0.434.

Table 2
National government health expenditure, social health expenditure and total health cost per capita, 2012–2019.

Year	Government health expenditure ($100 million)	Social health expenditure ($100 million)	Per capita health costs (yuan)
2012	8431.98	10,030.70	2068.8
2013	9545.81	11,393.79	2316.2
2014	10,579.23	13,437.75	2565.5
2015	12,475.28	16,506.71	2962.2
2016	13,910.31	19,096.68	3328.6
2017	15,205.87	22,258.81	3765.7
2018	16,399.13	25,810.78	4206.7
2019	18,016.95	29,150.57	4702.8

Table 3
Malmquist index of government and social healthcare inputs and outputs and their decomposition, 2012–2019.

Year	EFFCH		TECHCH		PECH		SECH		TFPCH	
	government	society	government	society	government	society	government	society	government	society
2012–2013	1.042	1.044	0.946	0.935	1.004	0.989	1.038	1.056	0.986	0.977
2013–2014	0.97	0.923	0.934	0.961	1.004	1.012	0.966	0.911	0.906	0.887
2014–2015	0.984	1.006	0.91	0.884	0.995	1.007	0.989	1	0.895	0.89
2015–2016	0.992	1.006	0.934	0.889	0.992	0.992	1	1.014	0.927	0.895
2016–2017	0.946	0.946	1.066	1.026	0.965	0.991	0.98	0.954	1.009	0.97
2017–2018	1.081	1.089	0.83	0.798	1.051	1.015	1.029	1.073	0.898	0.869
2018–2019	0.976	0.982	0.45	0.443	0.977	0.975	0.999	1.007	0.439	0.434

Overall, society's investment efficiency and technological progress in healthcare have fluctuated considerably. Despite this variability, there have been notable improvements in overall efficiency, particularly in terms of pure technical efficiency and scale efficiency. However, significant fluctuations in technological change and total factor productivity highlight challenges for society in sustaining consistent technological progress. In contrast, the government's performance in technical progress and total factor productivity has been relatively stable, though there have been some fluctuations and declines in certain years that warrant attention. This stability indicates a more consistent approach to technological advancements and productivity (Canavan et al., 2014), although there is room for improvement in addressing the noted variability (Charlton et al., 2015). The participation and contribution of social capital in healthcare are significant, enhancing the overall supply and diversity of healthcare services (Khanal et al., 2023). However, for more efficient and sustainable development, further advancements in technological progress and total factor productivity are essential (Prince et al., 2019). Continuous innovation and optimization in technology and efficiency are necessary to achieve more comprehensive improvements in the healthcare sector (Putrik et al., 2014). This suggests that while societal investments have positively impacted healthcare services, ongoing efforts are required to ensure these gains are both sustainable and progressive.

3.3. Trends in decomposition indicators over time

Figs. 1 and 2 illustrate the trends over time in total factor productivity and its decomposition indicators for government and social healthcare inputs and outputs. A comparative analysis of the total factor productivity of government healthcare input-output and social healthcare input-output in China shows that total factor productivity has fluctuated over time, especially during the 2017–2020 and 2018–2020 periods, when the decline was larger. It is also worth noting that the total factor productivity of social healthcare input output declined by 61.7 percent over the study period. The main reason for the decline is the significant drop in the rate of technical progress (TC). This suggests that, despite the huge inputs, the continued growth in per capita healthcare expenditure has placed a heavy burden on the population, and that the current allocation of resources and the efficiency of healthcare services need to be significantly improved in order to reduce the financial pressure on the population (Cho et al., 2018).

Fig. 3 compares the trends in total factor productivity (TFP) and its decompositions over time for government and social healthcare inputs and outputs. It can be found that social healthcare input output shows a trajectory of increasing technical progress efficiency, especially the growth of technical progress efficiency (TECHCH) exceeds the decline of technical efficiency (EFFCH), leading to an

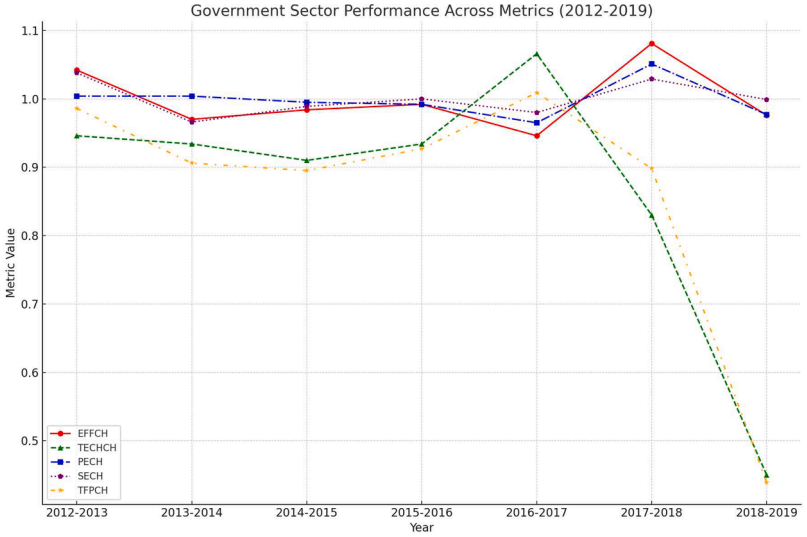


Fig. 1. Trends in government efficiency.

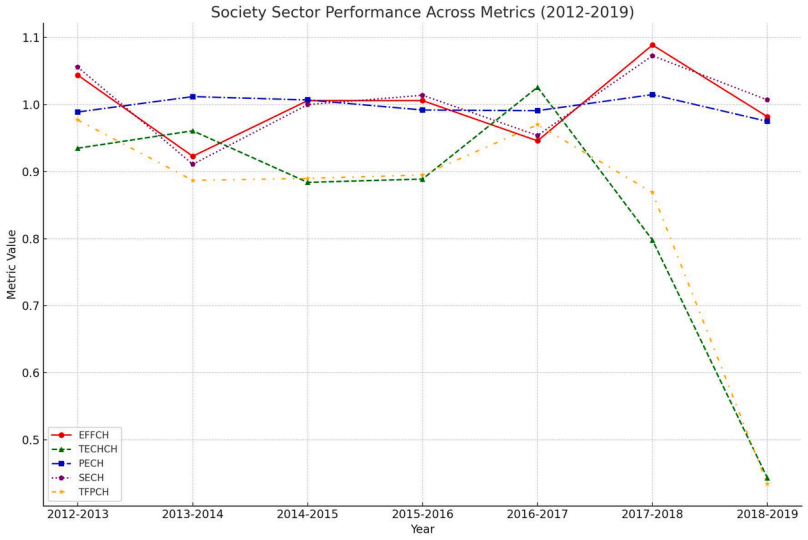


Fig. 2. Trends in social efficiency.

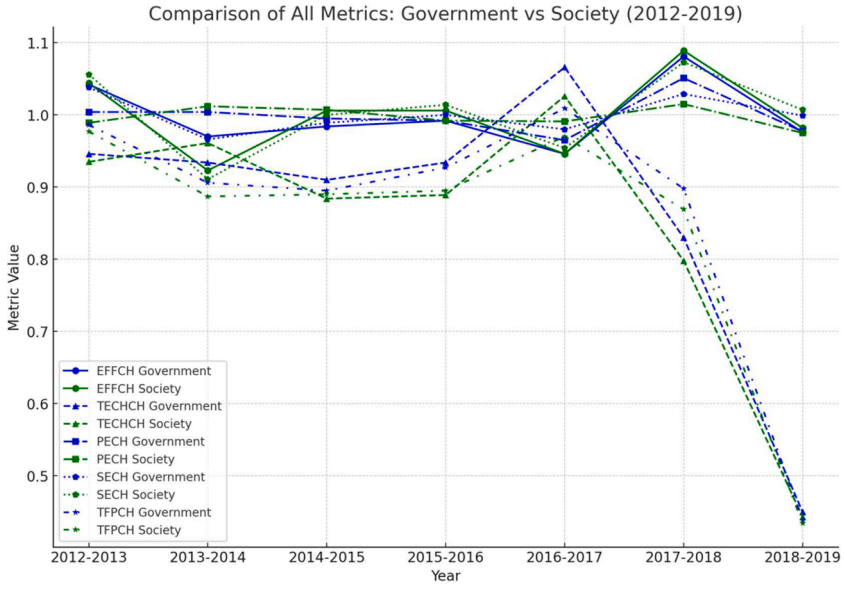


Fig. 3. Comparing government and societal efficiency.

upward trend in total factor productivity (TFP). This suggests that the increase in total factor productivity (TFP) of socio-medical inputs and outputs during this period stemmed mainly from the increase in the efficiency of technical progress.

3.4. Regional differences

According to the data in the Table 4, there are significant differences in the performance of social healthcare expenditures across various regions in China. The Efficiency Change Index (EFFCH) shows that Yunnan has the greatest improvement (1.032), while Shanghai has experienced a slight decline (0.985). In terms of the Technological Progress Index (TECHCH), Beijing (0.902), Tianjin (0.87), and Hebei (0.781) have relatively poor performance, whereas Yunnan (1.026), Qinghai (1.0), and Tibet (1.0) perform better. The Pure Technical Efficiency (PECH) is low in Tibet (0.845) and high in Tianjin (0.987), indicating that Tibet needs to invest more in training technical personnel and updating equipment. The Scale Efficiency (SECH) is higher in Liaoning (1.015), Jilin (1.013), and Inner Mongolia (1.005), while it is relatively low in Shanghai (0.986) and Fujian (0.994). The Total Factor Productivity Index (TFPCH) shows that the overall investment efficiency is low in Tibet (0.845), Guangxi (0.831), and Guizhou (0.785), but relatively stable in Beijing (0.877) and Tianjin (0.873).

Table 4
Input-output efficiencies of governmental and social health inputs in all regions of the country.

Region	Combined technical efficiency variation index		Technological progress index		Pure technical efficiency		Scale efficiency		Total factor productivity index	
	EFFCH		TECHCH		PECH		SECH		TFPCH	
	government	society	government	society	government	society	government	society	government	society
Beijing	0.979	0.972	0.865	0.902	1	1	0.979	0.972	0.846	0.877
Tianjin	1.01	1.003	0.857	0.87	0.989	0.987	1.022	1.016	0.866	0.873
Hebei	1	1	0.824	0.781	1	1	1	1	0.824	0.781
Shanxi	1	1	0.834	0.82	1	1	1	1	0.834	0.82
Inner Mongolia	1.02	1.005	0.838	0.806	1.008	1	1.012	1.005	0.855	0.81
Liaoning	1	1.002	0.817	0.83	1	0.986	1	1.015	0.817	0.832
Jilin	1.008	1.002	0.825	0.819	1.007	0.988	1.001	1.013	0.832	0.821
Heilongjiang	1	0.99	0.824	0.826	1	0.978	1	1.012	0.824	0.818
Shanghai	0.973	0.985	0.853	0.875	0.983	0.999	0.99	0.986	0.83	0.862
Jiangsu	0.989	0.993	0.844	0.833	1.008	1.008	0.982	0.985	0.835	0.827
Zhejiang	1.003	1.004	0.844	0.859	1.004	1.004	0.999	1.001	0.846	0.862
Anhui	1.011	1.009	0.842	0.821	1.012	1.002	1	1.006	0.852	0.829
Fujian	0.984	0.98	0.852	0.846	0.986	0.986	0.998	0.994	0.838	0.83
Jiangxi	0.98	0.986	0.847	0.777	0.969	0.987	1.011	0.998	0.83	0.766
Shandong	1	1.004	0.824	0.812	1	1	1	1.004	0.824	0.815
Henan	1	1	0.813	0.789	1	1	1	1	0.813	0.789
Hubei	0.987	0.982	0.84	0.823	0.99	0.986	0.997	0.995	0.829	0.808
Hunan	1	0.994	0.83	0.786	1	0.997	1	0.996	0.83	0.781
Guangdong	0.987	0.996	0.817	0.813	1	1	0.987	0.996	0.807	0.81
Guangxi	1	1	0.874	0.831	1	1	1	1	0.874	0.831
Hainan	0.987	0.989	0.868	0.859	0.983	0.991	1.004	0.998	0.856	0.849
Chongqing	1.011	1.012	0.842	0.829	1.009	1.005	1.002	1.007	0.851	0.839
Sichuan	0.99	0.99	0.812	0.788	1	1	0.99	0.99	0.804	0.78
Guizhou	1.007	0.996	0.857	0.788	0.995	0.998	1.013	0.998	0.863	0.785
Yunnan	1.027	1.032	0.829	0.767	1.01	1.026	1.016	1.006	0.851	0.791
Xizang	0.988	1	0.887	0.845	1	1	0.988	1	0.877	0.845
Shaanxi	1.023	1.018	0.848	0.845	1.005	1.003	1.018	1.015	0.868	0.86
Gansu	1.008	1.01	0.835	0.782	1	1	1.008	1.01	0.841	0.79
Qinghai	0.977	1.008	0.869	0.838	0.994	1	0.983	1.008	0.848	0.845
Ningxia	1	1	0.862	0.849	1	1	1	1	0.862	0.849
Xinjiang	0.993	0.986	0.825	0.822	0.993	0.984	1	1.002	0.819	0.81

Moreover, there are significant differences between social healthcare expenditures and government expenditures in terms of Efficiency Change, Technological Progress, Pure Technical Efficiency, Scale Efficiency, and Total Factor Productivity across different regions. In economically developed regions like Beijing, social expenditures outperform government expenditures in terms of technological progress and overall investment efficiency. Conversely, in less developed regions such as Hebei and Jilin, government expenditures perform better in terms of technological progress and overall investment efficiency. These differences reflect the varying needs and challenges in resource allocation, fund utilization, and technical personnel training across different regions.

In the future, policies and measures should be tailored to the specific conditions of each region to optimize resource allocation, enhance technological levels, and ensure that social healthcare expenditures more effectively serve society, thereby improving overall public health levels.

4. Conclusions

To reveal the significant challenges faced by social healthcare spending, this study investigates the efficiency of China's healthcare investments from 2012 to 2019. The findings indicate phenomena such as surplus medical resources, low efficiency in the utilization of medical funds, and diminishing returns due to continuous investments. Additionally, shortages in medical funds and healthcare technical personnel have significantly reduced the Total Factor Productivity Index (TFP), ultimately lowering overall investment efficiency. Although the scale efficiency of healthcare technology is improving, the Efficiency Change Index (EFFCH) reflects the combined effects of Scale Efficiency Change (SECH) and Pure Technical Efficiency Change (PECH). Moreover, the shortage of medical funds and technical personnel exposes the current healthcare system's inadequacies in responding to public health emergencies and managing chronic diseases. Insufficient funds and personnel make it difficult for hospitals and clinics to provide high-quality medical services, affecting patient outcomes and satisfaction, and consequently impacting the overall health level of society.

Future efforts must focus on enhancing investment levels at the societal level to ensure that healthcare expenditures lead to the rational allocation of medical resources. This approach will ensure that medical funds and technical personnel are more effectively utilized to serve society. By doing so, we can better meet the growing medical needs of the population and improve public health outcomes.

In conclusion, this study reveals the efficiency and current state of social investments in China's healthcare sector, highlighting the importance of focusing on social expenditures, particularly in the areas of resource allocation, fund utilization, and technical personnel

training. Only through coordinated efforts across various sectors can we achieve efficient utilization of medical resources, elevate the overall level of healthcare services, and ultimately promote societal health and harmonious development.

CRediT authorship contribution statement

Hongbo Li: Writing – original draft, Software, Resources, Investigation, Funding acquisition, Data curation, Conceptualization. **Xinyi He:** Writing – original draft, Methodology, Data curation. **Can Tan:** Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yaoyang Zhang:** Investigation, Formal analysis, Data curation, Conceptualization. **Dajun Yang:** Supervision, Formal analysis. **Yuan Li:** Writing – review & editing, Methodology, Investigation, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. This thesis was developed independently without the use of ChatGPT or any similar AI tools for content creation. ChatGPT's role was limited solely to grammar checks, ensuring linguistic precision.

Data availability

The authors do not have permission to share data.

Acknowledgments

This work is supported by the Doctoral Science Research Program at Xichang University "Research on the Influence of Job Insecurity on Enterprise Knowledge Workers' Innovative Behavior" (Grant/Award Number: YBS202321)

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